Growth Data Cleaning Report

Data Processing Methodology

This report summarises the results of a multi-stage cleaning pipeline designed to identify and correct anomalies in longitudinal tree growth data. The process includes local growth rate validation, robust z-score outlier detection within species-age cohorts, and linear interpolation for identified spike-revert anomalies.

Processing Metrics

Initial measurements: 983523
Flagged measurements to drop: 48085 / 982977 (4.89%)
Share of corrected points with >10% relative change: 0.06%
Flagged outlier measurements: 8826 / 933650 (0.95%)
Number of single scan trees being dropped: 343571
Final cleaned measurements: 581253
Total dropped measurements: 402270

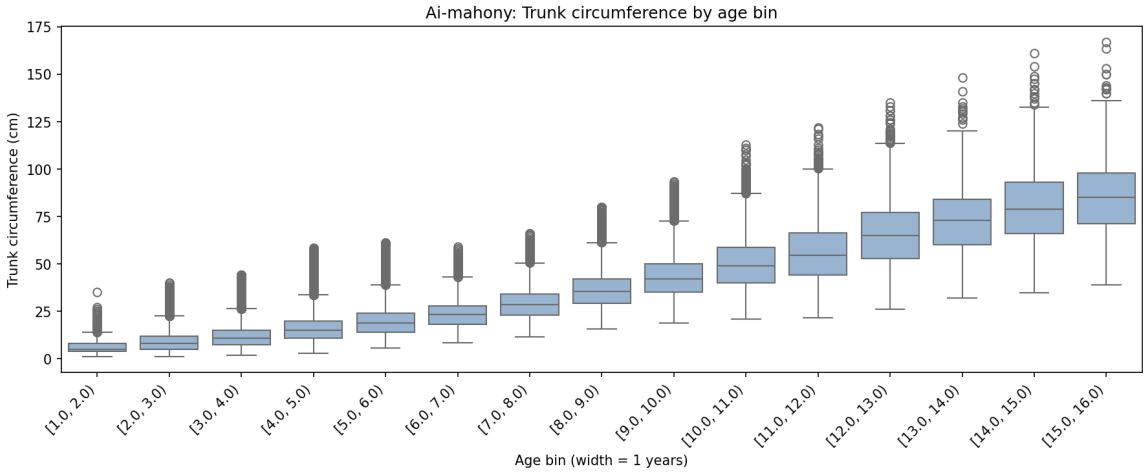
Cleaned Data Output

The resulting cleaned dataset provides a refined basis for growth modelling. Key outputs include *trunk_circumference_clean*, which represents the validated measurement, and *age_years_at_scan*, which normalises measurements to the time elapsed since planting.

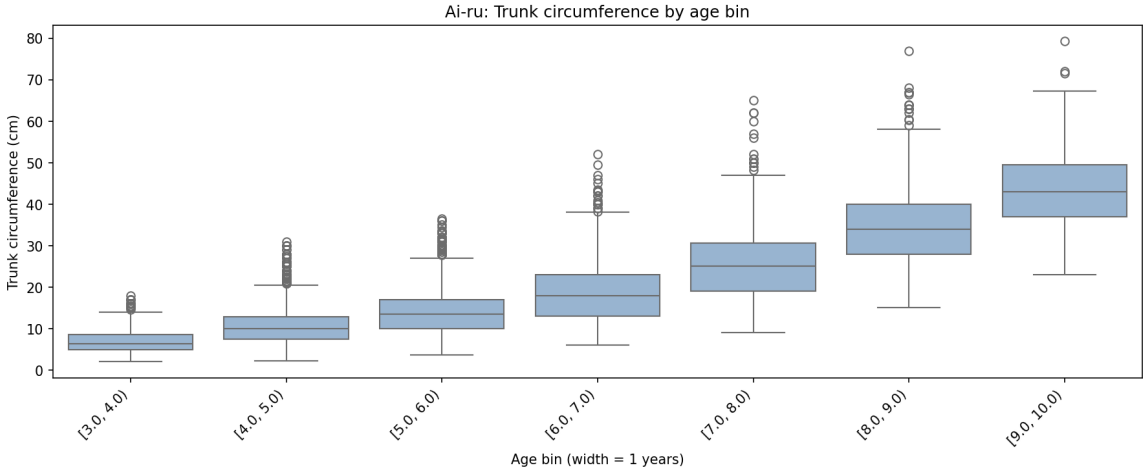
Interpreting Species Figures

The box plots on the following pages illustrate the distribution of trunk circumferences across different age bins (1-year intervals). The 'boxes' represent the interquartile range (IQR), the internal line represents the median, and the whiskers extend to 1.5x the IQR. Individual points (fliers) represent trees that are statistically significant but passed the cleaning thresholds.

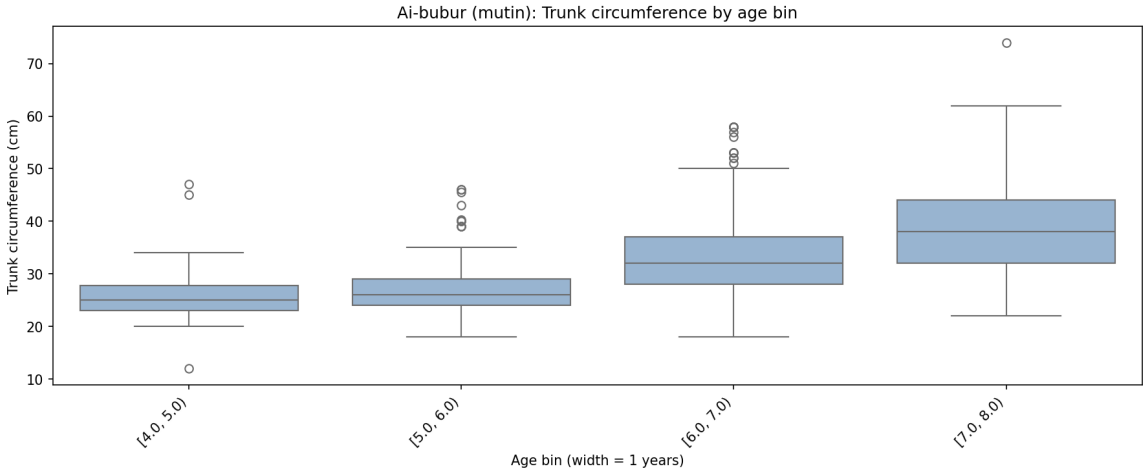
Species: Ai-mahony



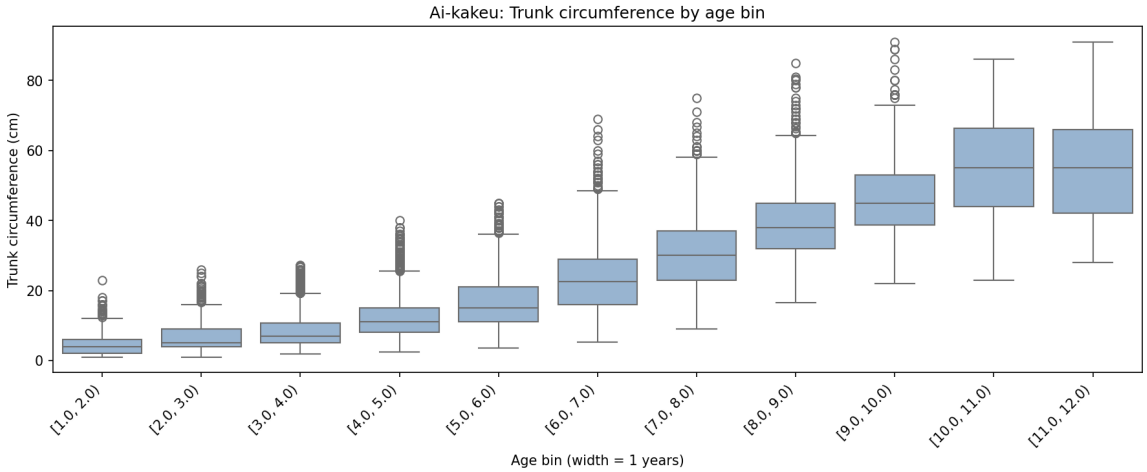
Species: Ai-ru



Species: Ai-bubur (mutin)



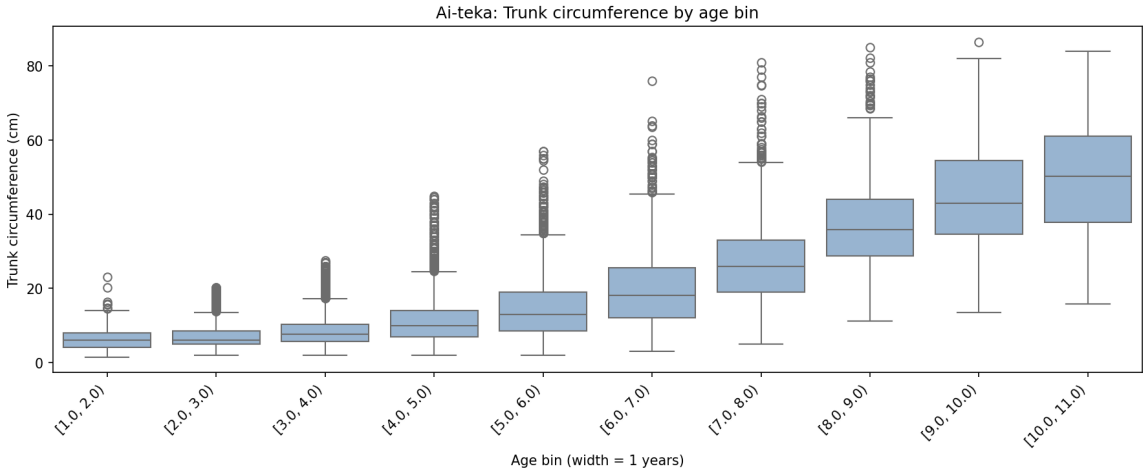
Species: Ai-kakeu



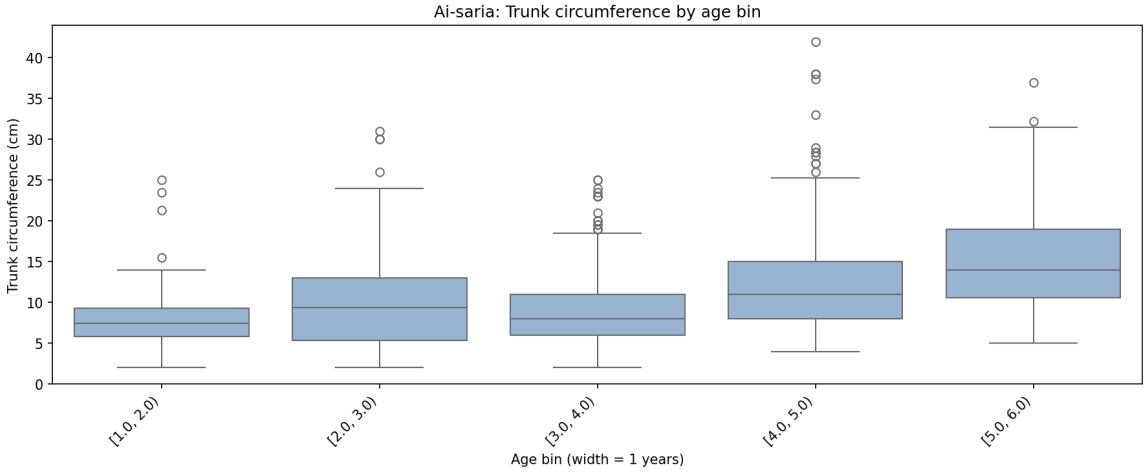
Species: Ai-na

Species: Ai-nitas

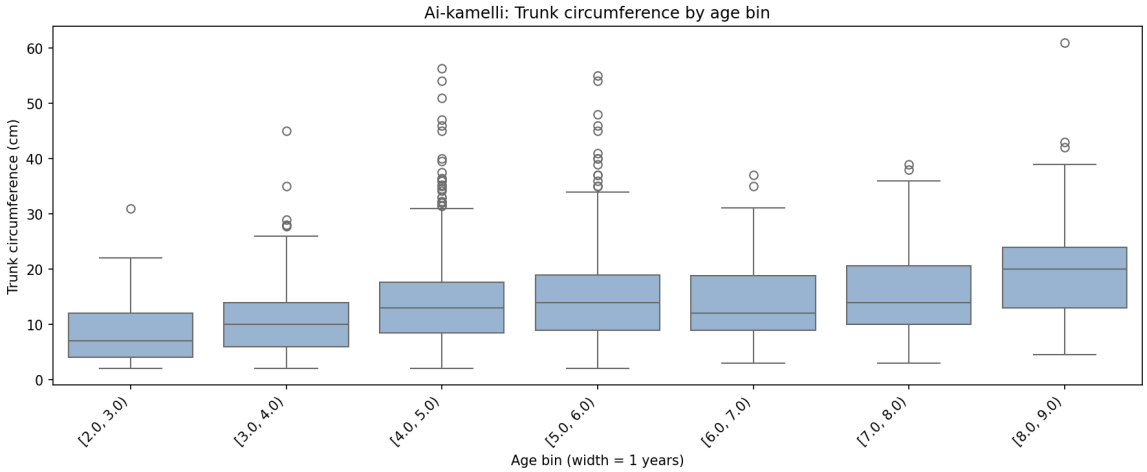
Species: Ai-teka

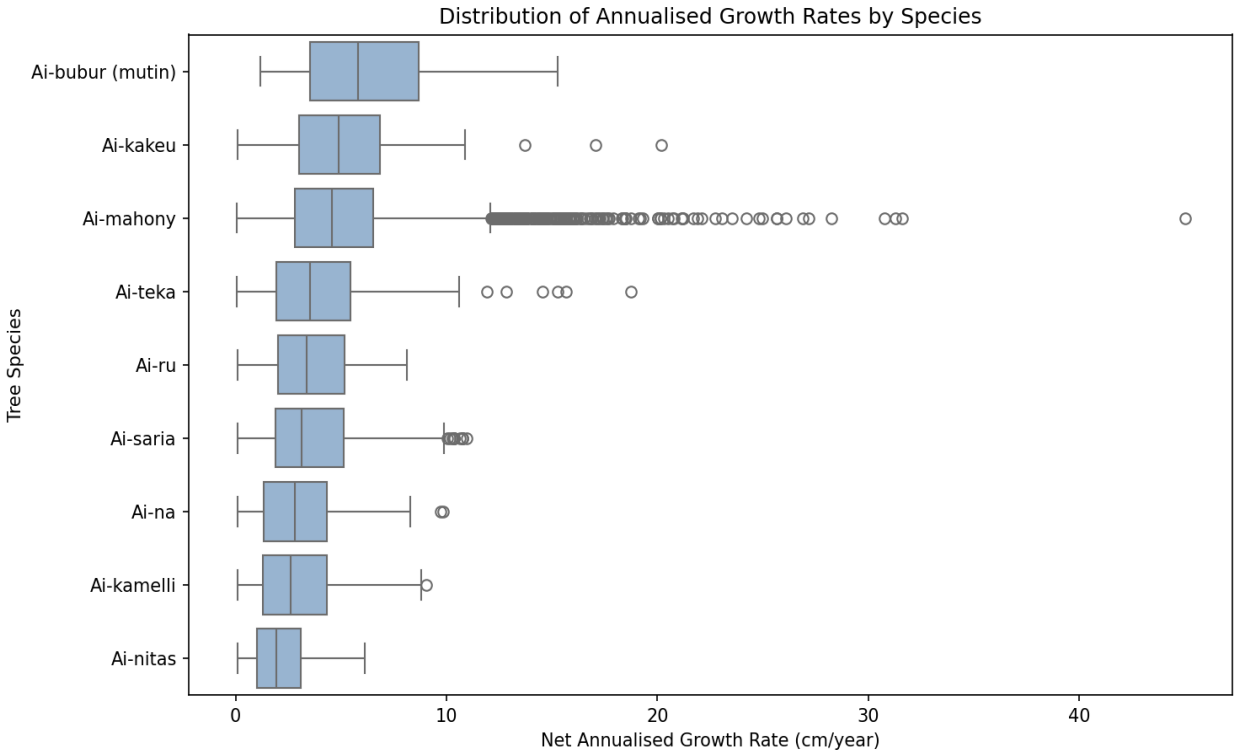


Species: Ai-saria



Species: Ai-kamelli



Overall Growth Rate Distributions

The chart below illustrates the annualised net growth rate (cm/year) calculated from the first and last valid scans of each tree, grouped by species.